Sushanth Buddhala

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San Ramon, California

PROFILE

Data Analytics Engineering graduate student with hands-on project experience building RAG pipelines, Agentic AI workflows and machine learning models. Skilled in Python, SQL, Power BI and fundamental of computer networks, with focus on designing prompt engineering workflows, implementing data validation, and translating complex datasets into actionable insights.

PROJECT EXPERIENCE

Compass - Agentic Multi-Agent Retail Analytics System *Feb 2026 – May 2026*

- Developed Compass to eliminate manual analyst bottlenecks in retail data workflows - architected a 5-agent pipeline (Planner, Data, Product, Supply Chain, Synthesizer) in Python using Gemini API function calling, Pydantic orchestrator, and DuckDB with Cursor AI, without LangChain.
- Built hybrid Flash/Pro fallback strategy, column synonym normalization, and prompt engineering across 5 system prompts to handle LLM variability; deployed Streamlit UI with Pandas, NumPy and Plotly charts with reasoning trace visualization.
- Designed a 40-case Pytest integration eval harness with 4-dimensional scoring, achieving 95% completion and 95.2% correctness across 3 bug-fix rounds surfacing 5 production bugs.

SARRDI - Dam Infrastructure Risk Analytics *Jan 2026 – May 2026*

- Built an ETL pipeline (Python, GeoPandas, AWS S3) integrating 5 datasets (NID, DCR, FEMA Fathom, USGS NHD, CDC SVI) across 2,712 dams, achieving 97.9% river-name completeness and 95% dam-to-water-feature mapping.
- Engineered a rule-based triage model (5 DAX-weighted dimensions: hazard, flood exposure, watershed cascade risk, social vulnerability, data quality), flagging 315 uninspected flood-zone dams and 74 missing Emergency Action Plans.
- Delivered an interactive Power BI dashboard with geospatial filtering across 2,712 dams, helping DCR prioritize inspections and surface 606 high-risk assets invisible in the National Inventory of Dams and save lives.

Housing Migration Analytics Platform (SQL + RAG) *Nov 2025 – Jan 2026*

- Developed an AI-powered analytics platform for non-SQL users exploring 395K+ U.S. housing and migration records via natural language, integrating Migration, AHS, and HUD CHAS datasets into a normalized SQLite database using Cursor AI.
- Built a schema-aware RAG pipeline (OpenAI embeddings, cosine similarity) converting natural-language questions into accurate, SELECT-only SQL, preventing hallucinations via schema validation and guardrails.
- Developed a FastAPI backend and Streamlit dashboard with interactive visualizations, SQL transparency, and schema validation, allowing users to safely analyze housing, migration, rent, and income trends without writing SQL.

TECHNICAL SKILLS

Languages: Python, SQL, DAX
Frameworks & Libraries: Pydantic, Streamlit, FastAPI, Pytest, Pandas, NumPy, Plotly, Scikit-learn, Tenacity
Databases & Platforms: SAP, Microsoft Excel, DuckDB, SQLite, AWS, Cursor AI, Snowflake, Power BI
Data & ML: Data Analysis, Data Modeling, Data Warehousing, Data Pipelines, ETL pipelines, Statistical Analysis, Machine Learning, Unsupervised learning
AI & LLM: Gemini API, OpenAI API, Function Calling, Prompt Engineering, Multi-Agent Systems, Workflow Orchestration, Model Evaluation, RAG Pipelines

EDUCATION

George Mason University Virginia, USA
M.S. in Data Analytics Engineering

- ***Relevant Coursework:*** *Analytics & Big Data, Intro to Analytics & Modeling, Principles of Data Management & Mining, Visualization for Analytics, Big Data Essentials, Data Mining for Business Analytics, Interpretable Machine Learning, Data Analytics Project*

Gokaraju Rangaraju Institute of Engineering and Technology Telangana, India
B.S. in Computer Science Engineering

- ***Relevant Coursework:*** *Linear Algebra Differential Calculus, Programming for Problem Solving, Differential Equation and Vector Calculus, Computer Networks, Database Management System, Python Programming, Artificial Intelligence, Data Analytics with Python, Machine Learning, Big Data Analytics, Cloud Computing, Data Warehousing and Data Mining*

PROFESSIONAL CERTIFICATES

• IBM Data Analyst Professional Certificate by Coursera • AWS Academy Graduate - Cloud Foundation • Generative AI for Data Analytics by Coursera